FUEL GAUGE

id092200032700

Purpose
- The fuel gauge notifies the driver of the fuel tank level.

Function
- The instrument cluster calculates the amount of fuel in the fuel tank based on the following signals and displays the fuel gauge segments.

Signal name	Sending module/part name	Communication method
Fuel gauge sender unit voltage signal	BCM	HS-CAN
Fuel injection amount signal	PCM	
Vehicle speed signal		

Fuel gauge flicker prevention function
- When the instrument cluster determines that the vehicle is being driven, it retards the response with which the calculated fuel tank level is reflected to the fuel gauge. As a result, fuel gauge flicker caused by the variation of fuel surface during driving is prevented.

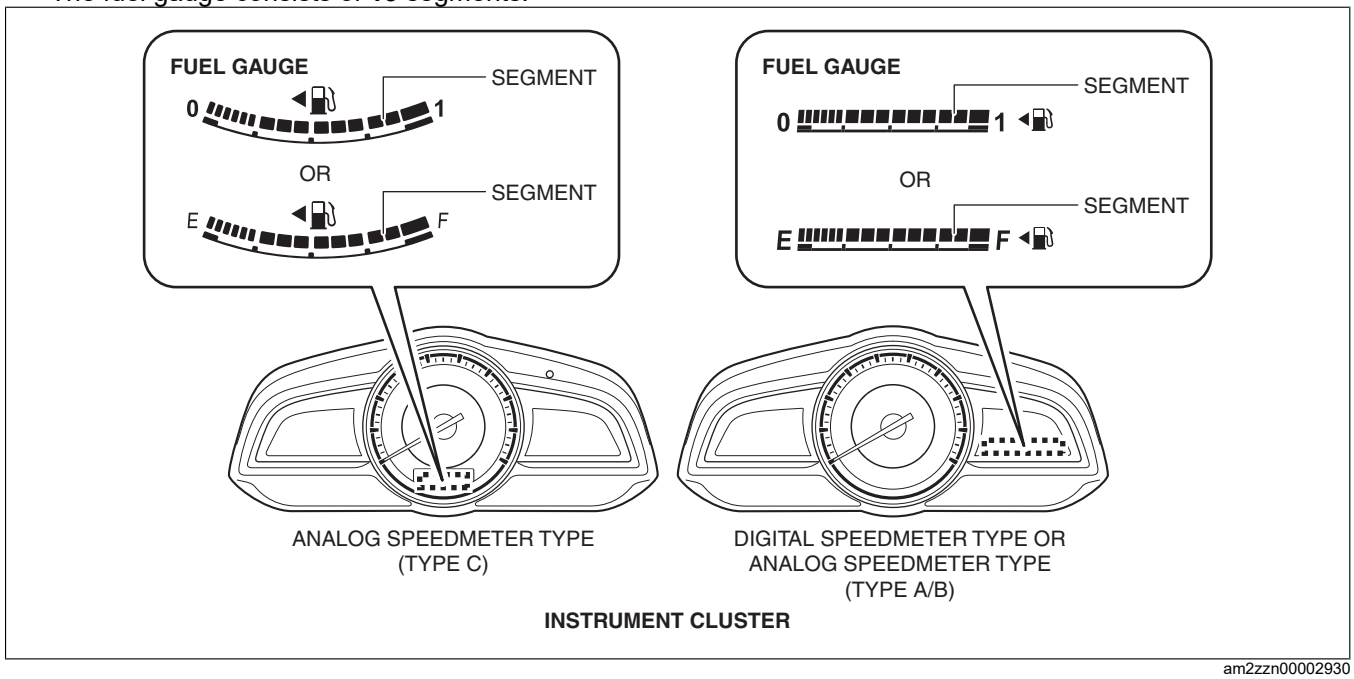
Refuel determination function
Analog speedometer type
- If the fuel tank level changes by **3 L {0.8 US gal, 0.7 Imp gal}** or **2 L {0.5 US gal, 0.4 Imp gal}** *1 **or more** while the vehicle is stopped, the instrument cluster determines that the vehicle is refueled.

*1 : **3 L {0.8 US gal, 0.7 Imp gal}** when the number of fuel gauge segments is **14 or less**, **2 L {0.5 US gal, 0.4 Imp gal}** when the number of fuel gauge segments is **15 or more**

Digital speedometer type
- If the fuel tank level changes by **5 L {1 US gal, 1 Imp gal}** or **2 L {0.5 US gal, 0.4 Imp gal}** *1 **or more** while the vehicle is stopped, the instrument cluster determines that the vehicle is refueled.

*1 : **5 L {1 US gal, 1 Imp gal}** when the number of fuel gauge segments is **13 or less**, **2 L {0.5 US gal, 0.4 Imp gal}** when the number of fuel gauge segments is **14 or more**

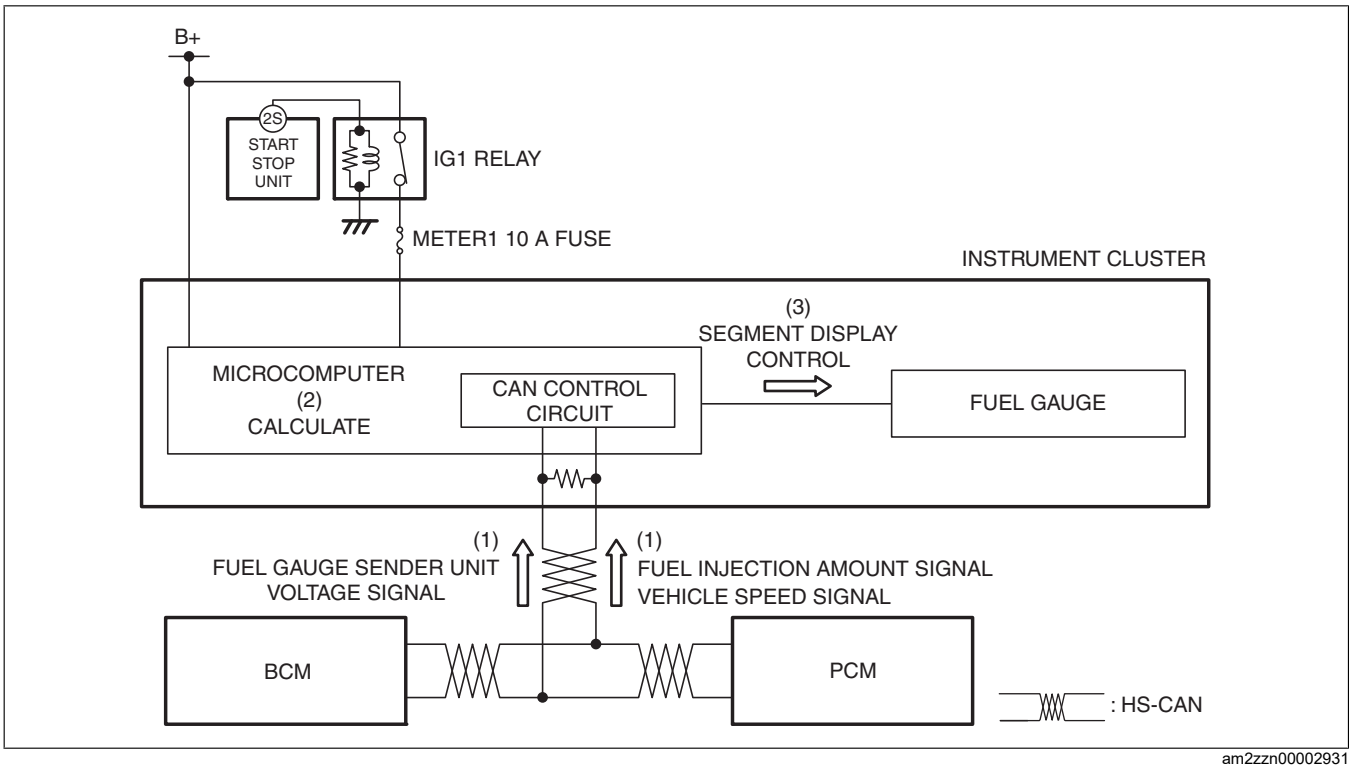
Construction
- The fuel gauge is displayed in the LCD of the instrument cluster.
- The fuel gauge consists of **15** segments.



INSTRUMENT CLUSTER

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Operation
1. When the ignition is switched ON (engine off or on), the instrument cluster receives (1) the fuel gauge sender unit voltage signal from the BCM, and the fuel injection amount signal and vehicle speed signal from the PCM via CAN communication.
2. The instrument cluster calculates the fuel tank level based on each signal (2).
3. The instrument cluster displays/turns off the fuel gauge segments according to the calculation result (3).



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Fail-safe

- Not applicable